

```
# =====  
# PURCHASE PATTERN ANALYSIS  
# =====  
  
# -----  
# 1. IMPORT REQUIRED LIBRARIES  
# -----  
  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns  
  
# -----  
# 2. LOAD CLEANED TRANSACTION DATA  
# -----  
  
# Load the cleaned transaction-level dataset created in Phase 2  
data = pd.read_csv("ecommerce_cleaned_transactions.csv")  
  
# Convert purchase date back to datetime format  
data["purchase_date"] = pd.to_datetime(  
    data["purchase_date"],  
    errors="coerce"  
)
```

```
# Display basic information
print("Dataset Shape:", data.shape)

print("\nFirst 5 Records:")
display(data.head())

# -----
# 3. BASIC PURCHASE STATISTICS
# -----

print("\nPurchase Statistics:")
display(
    data[
        [
            "product_price",
            "quantity",
            "total_purchase_amount"
        ]
    ].describe()
)
```

```
# -----  
# 4. TOTAL SALES / REVENUE  
# -----
```

```
total_revenue = data["total_purchase_amount"].sum()
```

```
print(  
    "Total Revenue:",  
    round(total_revenue, 2)  
)
```

```
# -----  
# 5. TOTAL NUMBER OF TRANSACTIONS  
# -----
```

```
total_transactions = data["customer_id"].count()
```

```
print(  
    "Total Transactions:",  
    total_transactions  
)
```

```
# -----  
# 6. TOTAL UNIQUE CUSTOMERS  
# -----
```

```
unique_customers = data["customer_id"].nunique()
```

```
print(  
    "Unique Customers:",  
    unique_customers  
)
```

```
# -----  
# 7. TOTAL QUANTITY SOLD  
# -----
```

```
total_quantity = data["quantity"].sum()
```

```
print(  
    "Total Quantity Sold:",  
    total_quantity  
)
```

```
# -----  
# 8. AVERAGE ORDER VALUE  
# -----
```

```
average_order_value = data[  
    "total_purchase_amount"  
].mean()
```

```
print(  
    "Average Transaction Value:",  
    round(average_order_value, 2)  
)
```

```
# =====  
# 9. PRODUCT CATEGORY ANALYSIS  
# =====
```

```
# -----  
# 9A. CATEGORY-WISE TRANSACTIONS  
# -----
```

```
category_transactions = (  
    data["product_category"]  
    .value_counts()  
    .reset_index()  
)
```

```
category_transactions.columns = [  
    "Product_Category",  
    "Transaction_Count"  
]  
  
print("\nTransactions by Product Category:")  
display(category_transactions)
```

```
# -----  
# 9B. CATEGORY-WISE REVENUE  
# -----
```

```
category_revenue = (  
    data  
    .groupby("product_category")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Quantity_Sold=("quantity", "sum"),  
        Total_Revenue=("total_purchase_amount", "sum"),  
        Average_Order_Value=("total_purchase_amount", "mean")  
    )  
    .reset_index()  
)
```

```
category_revenue = category_revenue.sort_values(  
    "Total_Revenue",  
    ascending=False  
)
```

```
print("\nProduct Category Performance:")  
display(category_revenue)
```

```
# -----  
# 9C. CATEGORY REVENUE PERCENTAGE  
# -----
```

```
category_revenue["Revenue_Percentage"] = (  
    category_revenue["Total_Revenue"]  
    / category_revenue["Total_Revenue"].sum()  
    * 100  
)
```

```
category_revenue = category_revenue.round(2)
```

```
print("\nCategory Revenue Contribution:")  
display(category_revenue)
```

```
# -----  
# 9B. CATEGORY-WISE REVENUE  
# -----  
  
category_revenue = (  
    data  
    .groupby("product_category")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Quantity_Sold=("quantity", "sum"),  
        Total_Revenue=("total_purchase_amount", "sum"),  
        Average_Order_Value=("total_purchase_amount", "mean")  
    )  
    .reset_index()  
)  
  
category_revenue = category_revenue.sort_values(  
    "Total_Revenue",  
    ascending=False  
)  
  
print("\nProduct Category Performance:")  
display(category_revenue)
```

```
# -----  
# 9C. CATEGORY REVENUE PERCENTAGE  
# -----
```

```
category_revenue["Revenue_Percentage"] = (  
    category_revenue["Total_Revenue"]  
    / category_revenue["Total_Revenue"].sum()  
    * 100  
)  
  
category_revenue = category_revenue.round(2)  
  
print("\nCategory Revenue Contribution:")  
display(category_revenue)
```

```
# -----  
# 9D. VISUALIZE CATEGORY REVENUE  
# -----
```

```
plt.figure(figsize=(6,3))  
  
sns.barplot(  
    data=category_revenue,  
    x="Total_Revenue",  
    y="product_category"  
)
```

```
plt.title("Revenue by Product Category")
plt.xlabel("Total Revenue")
plt.ylabel("Product Category")
plt.tight_layout()
plt.show()
```

```
# -----
# 9E. VISUALIZE CATEGORY QUANTITY
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(
    data=category_revenue.sort_values(
        "Quantity_Sold",
        ascending=False
    ),
    x="Quantity_Sold",
    y="product_category"
)
```

```
plt.title("Quantity Sold by Product Category")
plt.xlabel("Quantity Sold")
plt.ylabel("Product Category")
plt.tight_layout()
plt.show()
```

```
# -----  
# 10A. PAYMENT METHOD USAGE  
# -----
```

```
payment_usage = (  
    data["payment_method"]  
    .value_counts()  
    .reset_index()  
)
```

```
payment_usage.columns = [  
    "Payment_Method",  
    "Transaction_Count"  
]
```

```
print("\nPayment Method Usage:")  
display(payment_usage)
```

```
# -----  
# 10B. PAYMENT METHOD REVENUE  
# -----
```

```
payment_analysis = (  
    data  
    .groupby("payment_method")  
    .agg(  
        Transactions=("customer_id", "count"),
```

```
    Transactions=("customer_id", "count"),
    Total_Revenue=("total_purchase_amount", "sum"),
    Average_Order_Value=("total_purchase_amount", "mean")
)
.reset_index()
```

)

```
payment_analysis["Revenue_Percentage"] = (
    payment_analysis["Total_Revenue"]
    / payment_analysis["Total_Revenue"].sum()
    * 100
)
```

)

```
payment_analysis = payment_analysis.sort_values(
    "Total_Revenue",
    ascending=False
)
```

)

```
payment_analysis = payment_analysis.round(2)
```

```
print("\nPayment Method Analysis:")
display(payment_analysis)
```

```
# -----  
# 10C. PAYMENT METHOD REVENUE VISUALIZATION  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=payment_analysis,  
    x="Total_Revenue",  
    y="payment_method"  
)
```

```
plt.title("Revenue by Payment Method")  
plt.xlabel("Total Revenue")  
plt.ylabel("Payment Method")  
plt.tight_layout()  
plt.show()
```

```
# =====  
# 11. MONTHLY PURCHASE ANALYSIS  
# =====
```

```
# -----  
# 11A. CREATE YEAR AND MONTH  
# -----
```

```
data["purchase_year"] = (  
    data["purchase_date"].dt.year  
)
```

```
data["purchase_month"] = (  
    data["purchase_date"].dt.month  
)
```

```
data["year_month"] = (  
    data["purchase_date"]  
    .dt.to_period("M")  
    .astype(str)  
)
```

```
# 11B. MONTHLY SALES
```

```
# -----
```

```
monthly_sales = (  
    data  
    .groupby("year_month")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Unique_Customers=("customer_id", "nunique"),  
        Quantity_Sold=("quantity", "sum"),  
        Revenue=("total_purchase_amount", "sum")  
    )  
    .reset_index()  
)  
  
print("\nMonthly Purchase Analysis:")  
display(monthly_sales.head(20))
```

```
# -----
```

```
# 11C. MONTHLY REVENUE TREND
```

```
# -----
```

```
plt.figure(figsize=(10,5))
```

```
sns.lineplot(  
    data=monthly_sales,  
    x="year_month",
```

```
x="year_month",  
y="Revenue",  
marker="o"
```

```
)
```

```
plt.title("Monthly Revenue Trend")  
plt.xlabel("Month")  
plt.ylabel("Revenue")  
plt.xticks(rotation=90)  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 11D. MONTHLY TRANSACTION TREND  
# -----
```

```
plt.figure(figsize=(8,4))
```

```
sns.lineplot(  
    data=monthly_sales,  
    x="year_month",  
    y="Transactions",  
    marker="o"
```

```
)
```

```
plt.title("Monthly Transaction Trend")  
plt.xlabel("Month")  
plt.ylabel("Number of Transactions")
```

```
plt.xticks(rotation=90)
plt.tight_layout()
plt.show()
```

```
# =====
# 12. YEARLY PURCHASE ANALYSIS
# =====
```

```
yearly_sales = (
    data
    .groupby("purchase_year")
    .agg(
        Transactions=("customer_id", "count"),
        Unique_Customers=("customer_id", "nunique"),
        Quantity_Sold=("quantity", "sum"),
        Revenue=("total_purchase_amount", "sum")
    )
    .reset_index()
)

print("\nYearly Purchase Analysis:")
display(yearly_sales)
```

```
# -----  
# 12A. YEARLY REVENUE VISUALIZATION  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=yearly_sales,  
    x="purchase_year",  
    y="Revenue"  
)
```

```
plt.title("Yearly Revenue")  
plt.xlabel("Year")  
plt.ylabel("Revenue")  
plt.tight_layout()  
plt.show()
```

```
# 13. DAY-OF-WEEK PURCHASE PATTERN
```

```
# =====
```

```
# -----
```

```
# 13A. CREATE DAY OF WEEK
```

```
# -----
```

```
data["day_of_week"] = (  
    data["purchase_date"].dt.day_name()  
)
```

```
# Correct chronological order
```

```
day_order = [  
    "Monday",  
    "Tuesday",  
    "Wednesday",  
    "Thursday",  
    "Friday",  
    "Saturday",  
    "Sunday"  
]
```

```

# -----
# 13B. DAY-WISE ANALYSIS
# -----

day_analysis = (
    data
    .groupby("day_of_week")
    .agg(
        Transactions=("customer_id", "count"),
        Revenue=("total_purchase_amount", "sum"),
        Quantity_Sold=("quantity", "sum")
    )
    .reindex(day_order)
    .reset_index()
)

print("\nDay-of-Week Purchase Pattern:")
display(day_analysis)

# -----
# 13C. DAY-WISE TRANSACTIONS
# -----

plt.figure(figsize=(6,3))

sns.barplot(
    data=day_analysis,

```

```
x="day_of_week",  
y="Transactions"
```

```
)
```

```
plt.title("Transactions by Day of Week")  
plt.xlabel("Day")  
plt.ylabel("Number of Transactions")  
plt.xticks(rotation=30)  
plt.tight_layout()  
plt.show()
```

```
# -----  
# 13D. DAY-WISE REVENUE  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=day_analysis,  
    x="day_of_week",  
    y="Revenue"  
)
```

```
plt.title("Revenue by Day of Week")  
plt.xlabel("Day")
```

```
plt.ylabel("Revenue")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

```
# =====
# 14. HOURLY PURCHASE PATTERN
# =====
```

```
# -----
# 14A. EXTRACT PURCHASE HOUR
# -----
```

```
data["purchase_hour"] = (
    data["purchase_date"].dt.hour
)
```

```
# -----  
# 14B. HOURLY ANALYSIS  
# -----  
  
hourly_analysis = (  
    data  
    .groupby("purchase_hour")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Revenue=("total_purchase_amount", "sum"),  
        Quantity_Sold=("quantity", "sum")  
    )  
    .reset_index()  
)  
  
print("\nHourly Purchase Pattern:")  
display(hourly_analysis)  
  
# -----  
# 14C. HOURLY TRANSACTION TREND  
# -----  
  
plt.figure(figsize=(8,4))  
  
sns.lineplot(  
    data=hourly_analysis,  
    x="purchase_hour",
```

```
        y="Transactions",
        marker="o"
    )

plt.title("Transactions by Purchase Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Number of Transactions")
plt.xticks(range(24))
plt.tight_layout()
plt.show()

# -----
# 14D. HOURLY REVENUE TREND
# -----

plt.figure(figsize=(8,5))

sns.lineplot(
    data=hourly_analysis,
    x="purchase_hour",
    y="Revenue",
    marker="o"
)

plt.title("Revenue by Purchase Hour")
plt.xlabel("Hour of Day")
plt.ylabel("Revenue")
```

```
plt.xlabel("Hour of Day")
plt.ylabel("Revenue")
plt.xticks(range(24))
plt.tight_layout()
plt.show()
```

```
# =====
# 15. AGE GROUP ANALYSIS
# =====
```

```
# -----
# 15A. CREATE AGE GROUPS
# -----
```

```
data["age_group"] = pd.cut(
    data["customer_age"],
    bins=[0, 18, 25, 35, 45, 55, 65, 100],
    labels=[
        "Under 18",
        "18-25",
        "26-35",
        "36-45",
        "46-55",
        "56-65",
        "65+"
    ],
    right=True
)
```

```
# 15B. AGE GROUP PERFORMANCE
```

```
# -----
```

```
age_analysis = (  
    data  
    .groupby("age_group", observed=True)  
    .agg(  
        Transactions=("customer_id", "count"),  
        Unique_Customers=("customer_id", "nunique"),  
        Quantity_Sold=("quantity", "sum"),  
        Revenue=("total_purchase_amount", "sum"),  
        Average_Order_Value=("total_purchase_amount", "mean")  
    )  
    .reset_index()  
)
```

```
print("\nAge Group Purchase Analysis:")  
display(age_analysis)
```

```
# -----
```

```
# 15C. AGE GROUP REVENUE
```

```
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=age_analysis,
```

```
    x="age_group",
    y="Revenue"
)

plt.title("Revenue by Age Group")
plt.xlabel("Age Group")
plt.ylabel("Revenue")
plt.xticks(rotation=30)
plt.tight_layout()
plt.show()
```

```
# =====
# 16. GENDER-WISE PURCHASE ANALYSIS
# =====
```

```
gender_analysis = (
    data
    .groupby("gender")
    .agg(
        Transactions=("customer_id", "count"),
        Unique_Customers=("customer_id", "nunique"),
        Quantity_Sold=("quantity", "sum"),
        Revenue=("total_purchase_amount", "sum"),
        Average_Order_Value=("total_purchase_amount", "mean")
    )
    .reset_index()
)
```

```
print("\nGender-wise Purchase Analysis:")  
display(gender_analysis)
```

```
# -----  
# 16A. GENDER REVENUE VISUALIZATION  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=gender_analysis,  
    x="gender",  
    y="Revenue"  
)
```

```
plt.title("Revenue by Gender")  
plt.xlabel("Gender")  
plt.ylabel("Revenue")  
plt.tight_layout()  
plt.show()
```

```
# =====  
# 17. PRODUCT CATEGORY × GENDER ANALYSIS  
# =====
```

```
category_gender = (  
    data  
    .pivot_table(  
        index="product_category",  
        columns="gender",  
        values="total_purchase_amount",  
        aggfunc="sum"  
    )  
)  
  
print("\nRevenue by Product Category and Gender:")  
display(category_gender)
```

```
# -----  
# 17A. HEATMAP  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.heatmap(  
    category_gender,  
    annot=True,  
    fmt=".0f"
```

```

)

plt.title("Revenue: Product Category vs Gender")
plt.xlabel("Gender")
plt.ylabel("Product Category")
plt.tight_layout()
plt.show()

# =====
# 18. PRODUCT CATEGORY x AGE GROUP
# =====

category_age = (
    data
    .pivot_table(
        index="product_category",
        columns="age_group",
        values="total_purchase_amount",
        aggfunc="sum",
        observed=False
    )
)

print("\nRevenue by Product Category and Age Group:")
display(category_age)

```

```

# -----
# 18A. HEATMAP
# -----

plt.figure(figsize=(8,4))

sns.heatmap(
    category_age,
    annot=True,
    fmt=".0f"
)

plt.title("Revenue: Product Category vs Age Group")
plt.xlabel("Age Group")
plt.ylabel("Product Category")
plt.tight_layout()
plt.show()

# =====
# 19. PRODUCT CATEGORY x PAYMENT METHOD
# =====

category_payment = (
    data
    .pivot_table(
        index="product_category",
        columns="payment_method",

```

```
        values="total_purchase_amount",
        aggfunc="sum"
    )
)

print("\nRevenue by Product Category and Payment Method:")
display(category_payment)
```

```
# -----
# 19A. HEATMAP
# -----
```

```
plt.figure(figsize=(8,4))

sns.heatmap(
    category_payment,
    annot=True,
    fmt=".0f"
)

plt.title("Revenue: Product Category vs Payment Method")
plt.xlabel("Payment Method")
plt.ylabel("Product Category")
plt.tight_layout()
plt.show()
```

```
# =====  
# 20. PURCHASE QUANTITY ANALYSIS  
# =====
```

```
quantity_analysis = (  
    data  
    .groupby("quantity")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Revenue=("total_purchase_amount", "sum")  
    )  
    .reset_index()  
)
```

```
print("\nQuantity-wise Purchase Analysis:")  
display(quantity_analysis)
```

```
# -----  
# 20A. QUANTITY DISTRIBUTION  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=quantity_analysis,  
    x="quantity",  
    y="Transactions"
```

```

)

plt.title("Transactions by Quantity Purchased")
plt.xlabel("Quantity")
plt.ylabel("Transactions")
plt.tight_layout()
plt.show()

# =====
# 21. PRICE RANGE ANALYSIS
# =====

# -----
# 21A. CREATE PRODUCT PRICE BANDS
# -----

data["price_range"] = pd.cut(
    data["product_price"],
    bins=[0, 100, 500, 1000, 2500, 5000, np.inf],
    labels=[
        "₹0-100",
        "₹101-500",
        "₹501-1000",
        "₹1001-2500",
        "₹2501-5000",
        "₹5000+"
    ]
]

```

)

```
# -----  
# 21B. PRICE RANGE PERFORMANCE  
# -----
```

```
price_analysis = (  
    data  
    .groupby("price_range", observed=True)  
    .agg(  
        Transactions=("customer_id", "count"),  
        Quantity_Sold=("quantity", "sum"),  
        Revenue=("total_purchase_amount", "sum")  
    )  
    .reset_index()  
)
```

```
print("\nPrice Range Analysis:")  
display(price_analysis)
```

```
# -----  
# 21C. PRICE RANGE REVENUE  
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=price_analysis,  
    x="price_range",  
    y="Revenue"  
)
```

```
plt.title("Revenue by Product Price Range")  
plt.xlabel("Price Range")  
plt.ylabel("Revenue")  
plt.xticks(rotation=30)  
plt.tight_layout()  
plt.show()
```

```

# =====
# 22. RETURN / REFUND PATTERN
# =====

# -----
# 22A. RETURN STATUS
# -----

# Convert Returns into a binary return indicator
data["return_flag"] = (
    data["returns"]
    .fillna(0)
    .astype(int)
)

return_analysis = (
    data
    .groupby("return_flag")
    .agg(
        Transactions=("customer_id", "count"),
        Revenue=("total_purchase_amount", "sum")
    )
    .reset_index()
)

print("\nReturn Analysis:")
display(return_analysis)

```

```
# 22B. OVERALL RETURN RATE
```

```
# -----
```

```
return_rate = (  
    data["return_flag"].mean() * 100  
)
```

```
print(  
    "Overall Return Rate:",  
    round(return_rate, 2),  
    "%"  
)
```

```
# -----
```

```
# 22C. RETURN RATE BY CATEGORY
```

```
# -----
```

```
category_returns = (  
    data  
    .groupby("product_category")  
    .agg(  
        Transactions=("customer_id", "count"),  
        Returns=("return_flag", "sum"),  
        Return_Rate=("return_flag", "mean")  
    )  
    .reset_index()  
)
```

```
category_returns = category_returns.round(2)
```

```
print("\nReturn Rate by Product Category:")
```

```
display(category_returns)
```

```
# -----
```

```
# 22D. RETURN RATE VISUALIZATION
```

```
# -----
```

```
plt.figure(figsize=(6,3))
```

```
sns.barplot(  
    data=category_returns,  
    x="Return_Rate",  
    y="product_category"  
)
```

```
plt.title("Return Rate by Product Category")
```

```
plt.xlabel("Return Rate (%)")
```

```
plt.ylabel("Product Category")
```

```
plt.tight_layout()
```

```
plt.show()
```

```
# =====  
# 23. TOP CUSTOMERS BY TOTAL SPENDING  
# =====
```

```
top_customers = (  
    data  
    .groupby(  
        ["customer_id", "customer_name"],  
        as_index=False  
    )  
    .agg(  
        Total_Spending=("total_purchase_amount", "sum"),  
        Purchase_Frequency=("customer_id", "count"),  
        Total_Quantity=("quantity", "sum")  
    )  
    .sort_values(  
        "Total_Spending",  
        ascending=False  
    )  
)  
  
print("\nTop 20 Customers by Spending:")  
display(  
    top_customers.head(20)  
)
```

```
# =====  
# 24. TOP CUSTOMERS BY PURCHASE FREQUENCY  
# =====
```

```
frequent_customers = (  
    top_customers  
    .sort_values(  
        "Purchase_Frequency",  
        ascending=False  
    )  
)  
  
print("\nTop 20 Customers by Purchase Frequency:")  
display(  
    frequent_customers.head(20)  
)
```

```
# =====  
# 25. CATEGORY-CUSTOMER PURCHASE MATRIX  
# =====
```

```
customer_category = (  
    data  
    .pivot_table(  
        index="customer_id",  
        columns="product_category",
```

```

        values="total_purchase_amount",
        aggfunc="sum",
        fill_value=0
    )
)

print("\nCustomer x Product Category Matrix:")
display(
    customer_category.head()
)

# =====
# 26. FIND MOST POPULAR CATEGORY
# =====

most_popular_category = (
    data["product_category"]
    .value_counts()
    .idxmax()
)

print(
    "Most Popular Product Category:",
    most_popular_category
)

```

```
# =====  
# 27. FIND HIGHEST-REVENUE CATEGORY  
# =====
```

```
highest_revenue_category = (  
    category_revenue  
    .sort_values(  
        "Total_Revenue",  
        ascending=False  
    )  
    .iloc[0]["product_category"]  
)
```

```
print(  
    "Highest Revenue Category:",  
    highest_revenue_category  
)
```

```
# =====  
# 28. FIND MOST USED PAYMENT METHOD  
# =====
```

```
most_used_payment = (  
    data["payment_method"]  
    .value_counts()  
    .idxmax()  
)
```

```
print(
    "Most Used Payment Method:",
    most_used_payment
)

# =====
# 29. FIND PEAK PURCHASE DAY
# =====

peak_purchase_day = (
    day_analysis
    .sort_values(
        "Transactions",
        ascending=False
    )
    .iloc[0]["day_of_week"]
)

print(
    "Peak Purchase Day:",
    peak_purchase_day
)
```

```
# =====  
# 30. FIND PEAK PURCHASE HOUR  
# =====
```

```
peak_purchase_hour = (  
    hourly_analysis  
        .sort_values(  
            "Transactions",  
            ascending=False  
        )  
        .iloc[0]["purchase_hour"]  
)
```

```
print(  
    "Peak Purchase Hour:",  
    peak_purchase_hour  
)
```

```
# =====  
# 31. FIND BEST REVENUE MONTH  
# =====
```

```
best_revenue_month = (  
    monthly_sales  
        .sort_values(  
            "Revenue",
```

```

        "Revenue",
        ascending=False
    )
    .iloc[0]
)

print(
    "Highest Revenue Month:",
    best_revenue_month["year_month"]
)

print(
    "Revenue:",
    round(best_revenue_month["Revenue"], 2)
)

# =====
# 32. SAVE PHASE 4 ANALYSIS FILES
# =====

# Save category analysis
category_revenue.to_csv(
    "category_purchase_analysis.csv",
    index=False
)

# Save payment analysis
payment_analysis.to_csv(

```

```
"monthly_purchase_analysis.csv",
```

```
index=False
```

```
)
```

```
# Save yearly analysis
```

```
yearly_sales.to_csv(
```

```
"yearly_purchase_analysis.csv",
```

```
index=False
```

```
)
```

```
# Save day analysis
```

```
day_analysis.to_csv(
```

```
"day_of_week_analysis.csv",
```

```
index=False
```

```
)
```

```
# Save hourly analysis
```

```
hourly_analysis.to_csv(
```

```
"hourly_purchase_analysis.csv",
```

```
index=False
```

```
)
```

```
# Save age analysis
```

```
age_analysis.to_csv(
```

```
"age_group_purchase_analysis.csv",
```

```
index=False
```

```
)
```

```
# Save gender analysis
gender_analysis.to_csv(
    "gender_purchase_analysis.csv",
    index=False
)

# Save return analysis
category_returns.to_csv(
    "category_return_analysis.csv",
    index=False
)

# Save top customers
top_customers.to_csv(
    "top_customers_analysis.csv",
    index=False
)

# =====
# 33. SAVE ENRICHED TRANSACTION DATA
# =====

# Save the transaction dataset with the newly created
# purchase-pattern features
data.to_csv(
    "purchase_pattern_analysis_data.csv",
```

```
"purchase_pattern_analysis_data.csv",
```

```
index=False
```

```
)
```

```
# =====
```

```
# 34. FINAL STEP 4 SUMMARY
```

```
# =====
```

```
print("\n" + "=" * 60)
```

```
print("STEP 4: PURCHASE PATTERN ANALYSIS COMPLETED")
```

```
print("=" * 60)
```

```
print("\nKey Analysis Completed:")
```

```
print("✓ Overall purchase statistics")
```

```
print("✓ Product category analysis")
```

```
print("✓ Payment method analysis")
```

```
print("✓ Monthly purchase trends")
```

```
print("✓ Yearly purchase trends")
```

```
print("✓ Day-of-week purchase patterns")
```

```
print("✓ Hourly purchase patterns")
```

```
print("✓ Age group analysis")
```

```
print("✓ Gender-wise analysis")
```

```
print("✓ Category × Gender analysis")
```

```
print("✓ Category × Age analysis")
```

```
print("✓ Category × Payment analysis")
```

```
print("✓ Quantity analysis")
```

```
print("✓ Product price analysis")
```

```
print("✓ Return analysis")
```

```
print("✓ Top customer analysis")
```

```
print("\nImportant Findings:")
```

```
print(  
    "Most Popular Category:",  
    most_popular_category  
)
```

```
print(  
    "Highest Revenue Category:",  
    highest_revenue_category  
)
```

```
print(  
    "Most Used Payment Method:",  
    most_used_payment  
)
```

```
print(  
    "Peak Purchase Day:",  
    peak_purchase_day  
)
```

```
print(  
    "Peak Purchase Hour:",  
    peak_purchase_hour  
)
```

```
print(
    "Overall Return Rate:",
    round(return_rate, 2),
    "%"
)

print("\nOutput files created:")
print("1. category_purchase_analysis.csv")
print("2. payment_method_analysis.csv")
print("3. monthly_purchase_analysis.csv")
print("4. yearly_purchase_analysis.csv")
print("5. day_of_week_analysis.csv")
print("6. hourly_purchase_analysis.csv")
print("7. age_group_purchase_analysis.csv")
print("8. gender_purchase_analysis.csv")
print("9. category_return_analysis.csv")
print("10. top_customers_analysis.csv")
print("11. purchase_pattern_analysis_data.csv")
```

Dataset Shape: (250000, 17)

First 5 Records:

	customer_id	purchase_date	product_category	product_price	quantity	total_purchase_amount	payment_method	customer_age	returns	customer_name	gender	c
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christine Hernandez	Male	
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christine Hernandez	Male	
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	37	0.0	Christine Hernandez	Male	
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christine Hernandez	Male	
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	James Grant	Female	

Purchase Statistics:

	product_price	quantity	total_purchase_amount
count	250000.000000	250000.000000	250000.000000
mean	254.659512	2.998896	2725.370732
std	141.568577	1.414694	1442.933565
min	10.000000	1.000000	100.000000
25%	132.000000	2.000000	1477.000000
50%	255.000000	3.000000	2724.000000
75%	377.000000	4.000000	3974.000000
max	500.000000	5.000000	5350.000000

Total Revenue: 681342683
Total Transactions: 250000
Unique Customers: 49673
Total Quantity Sold: 749724
Average Transaction Value: 2725.37

Transactions by Product Category:

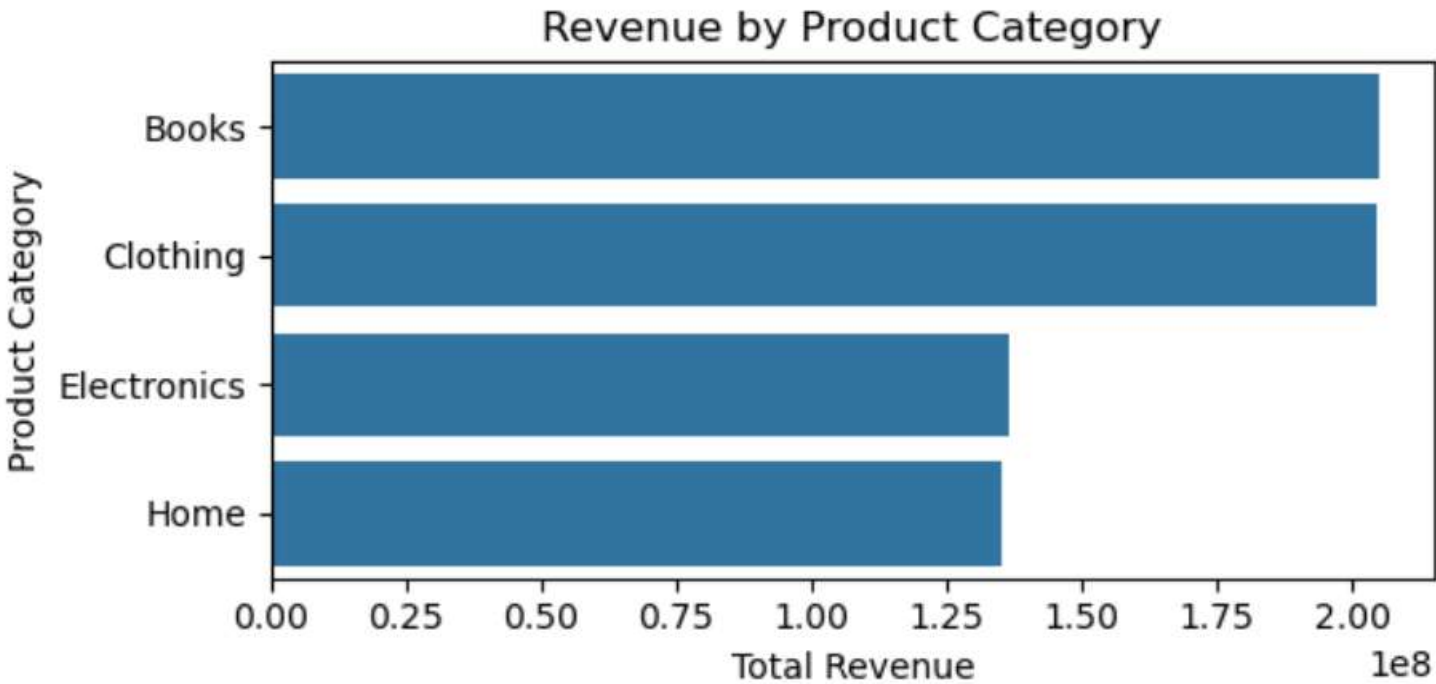
	Product_Category	Transaction_Count
0	Clothing	75052
1	Books	74912
2	Electronics	50185
3	Home	49851

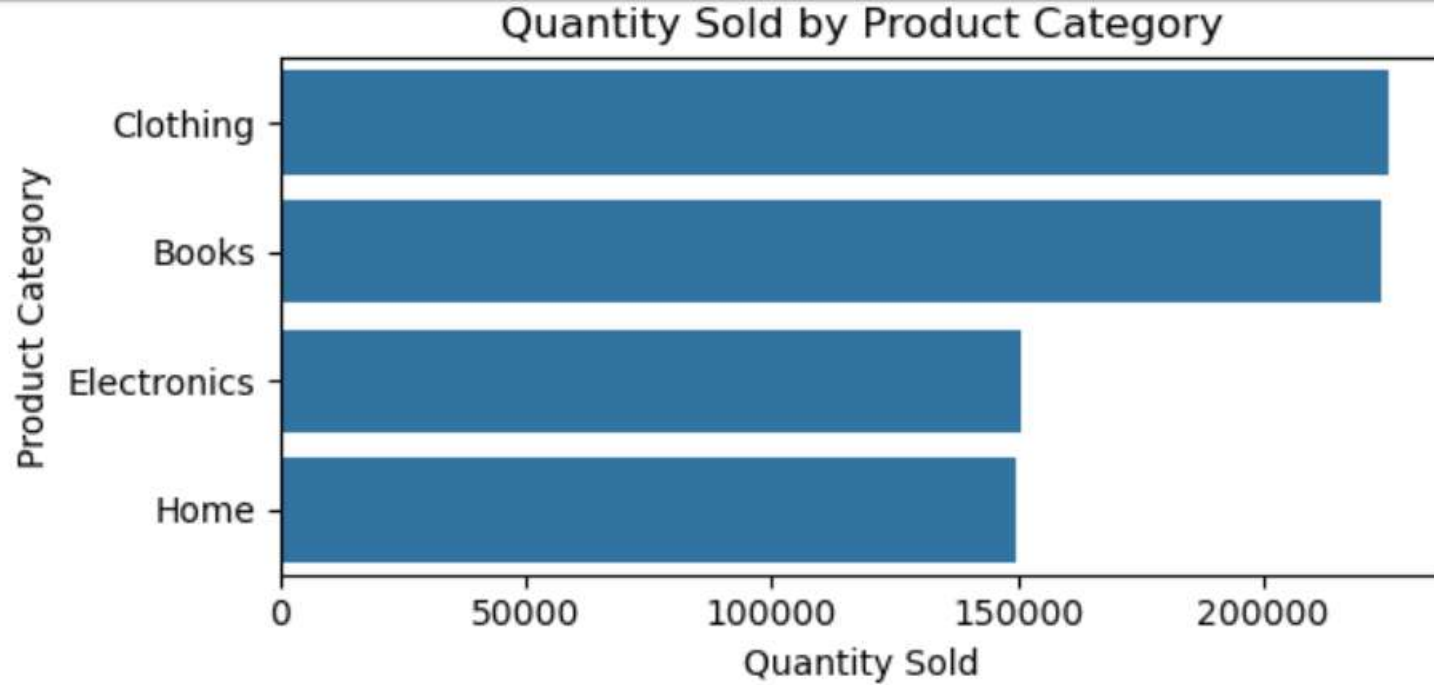
Product Category Performance:

	product_category	Transactions	Quantity_Sold	Total_Revenue	Average_Order_Value
0	Books	74912	223876	204939601	2735.737946
1	Clothing	75052	225322	204532405	2725.209255
2	Electronics	50185	150828	136599467	2721.918243
3	Home	49851	149698	135271210	2713.510461

Category Revenue Contribution:

	product_category	Transactions	Quantity_Sold	Total_Revenue	Average_Order_Value	Revenue_Percentage
0	Books	74912	223876	204939601	2735.74	30.08
1	Clothing	75052	225322	204532405	2725.21	30.02
2	Electronics	50185	150828	136599467	2721.92	20.05
3	Home	49851	149698	135271210	2713.51	19.85



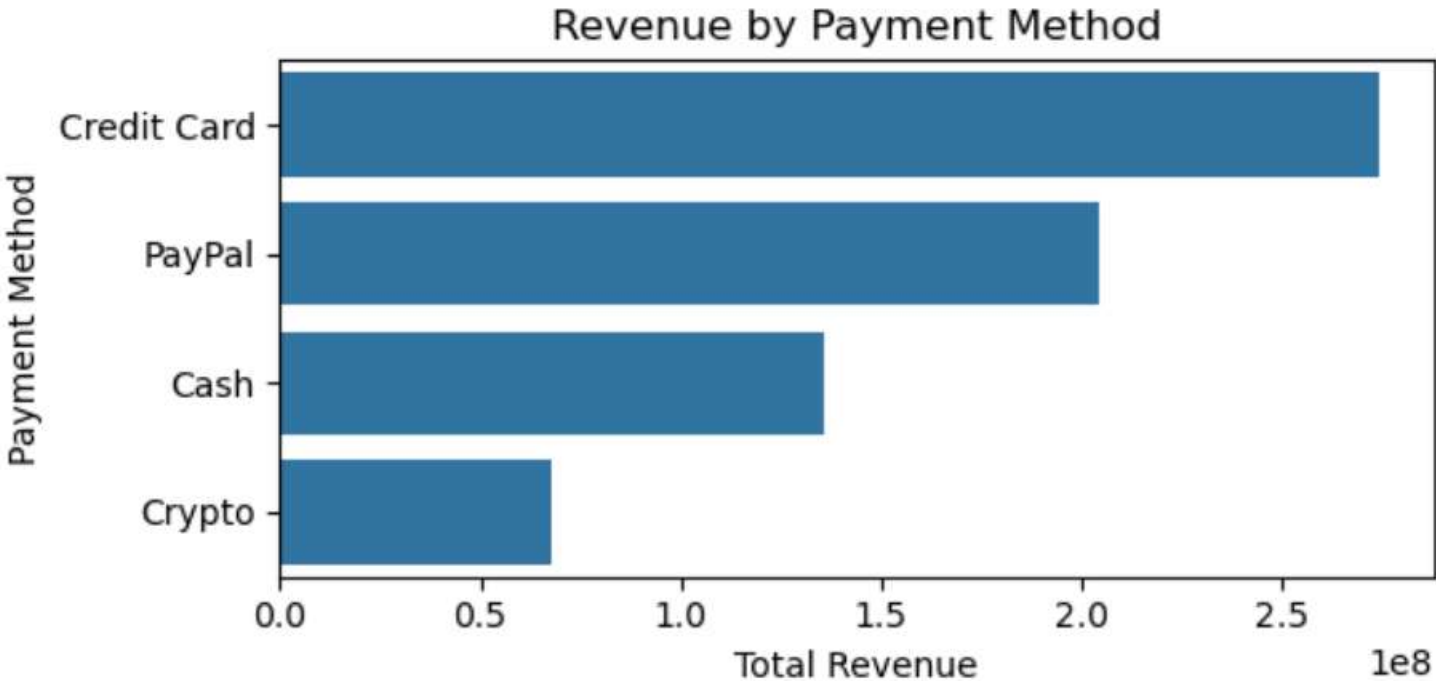


Payment Method Usage:

	Payment_Method	Transaction_Count
0	Credit Card	100486
1	PayPal	74837
2	Cash	49894
3	Crypto	24783

Payment Method Analysis:

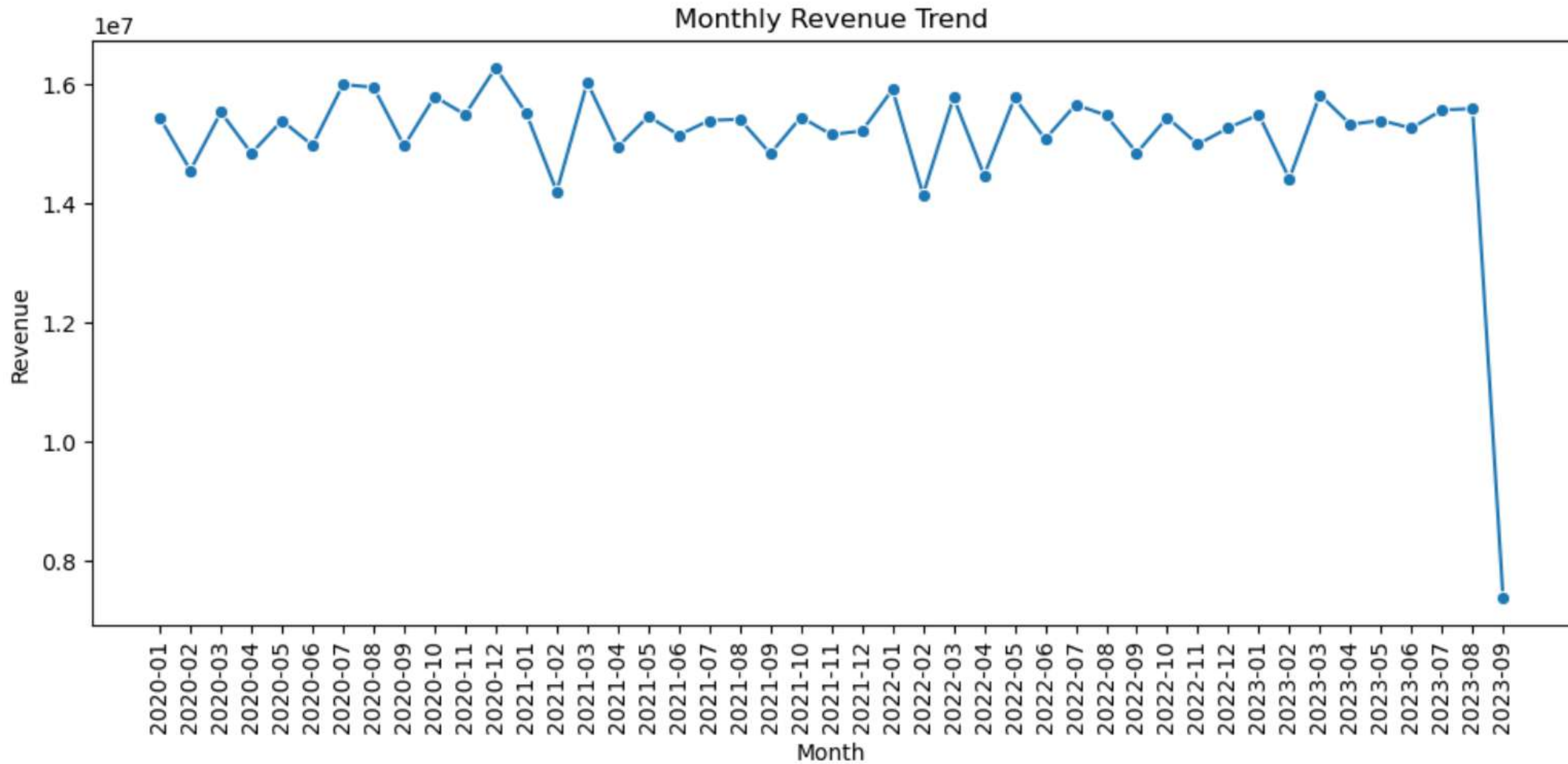
	payment_method	Transactions	Total_Revenue	Average_Order_Value	Revenue_Percentage
1	Credit Card	100486	274152396	2728.26	40.24
3	PayPal	74837	204026613	2726.28	29.94
0	Cash	49894	135705551	2719.88	19.92
2	Crypto	24783	67458123	2721.95	9.90



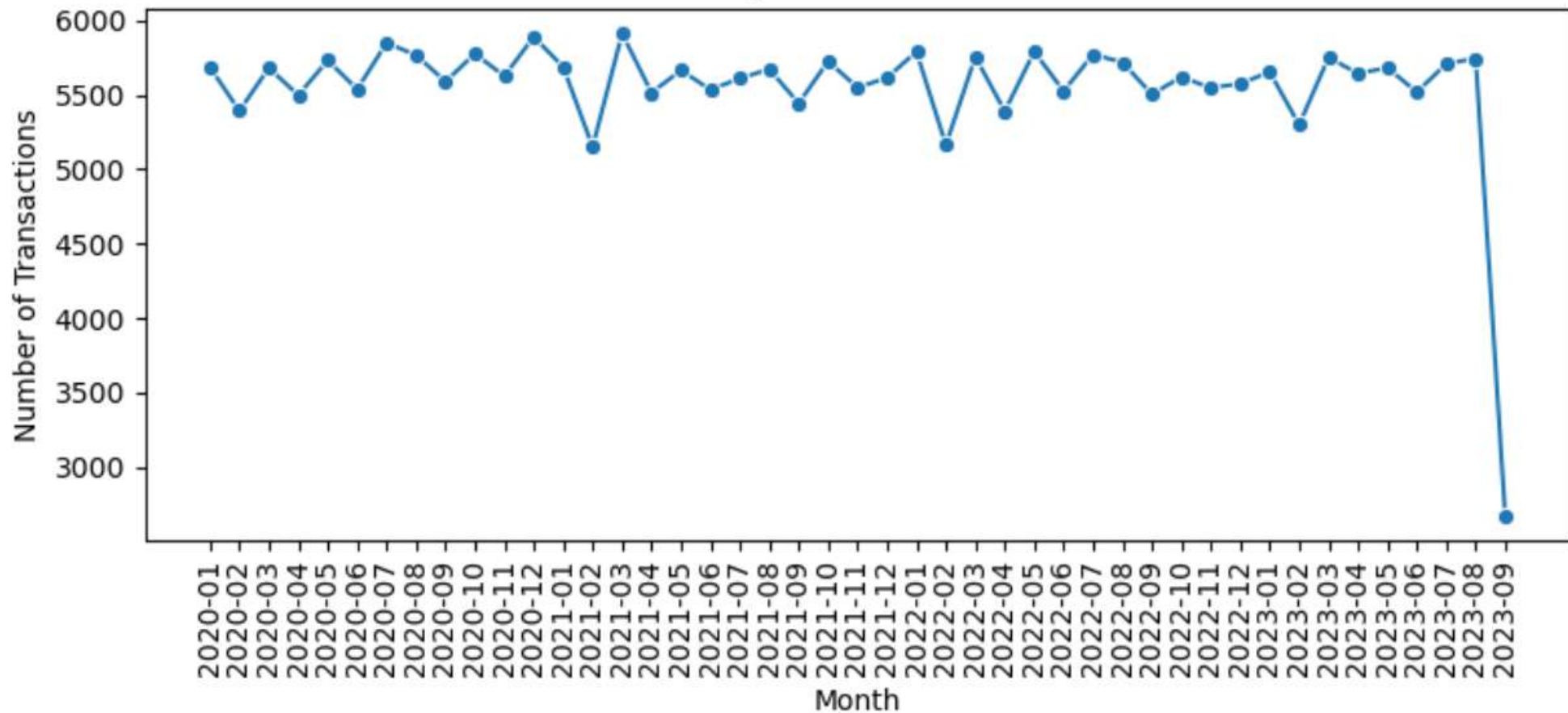
Monthly Purchase Analysis:

	year_month	Transactions	Unique_Customers	Quantity_Sold	Revenue
0	2020-01	5687	5376	17256	15442459
1	2020-02	5394	5099	16172	14561593
2	2020-03	5683	5365	17038	15537311
3	2020-04	5499	5193	16451	14847491
4	2020-05	5734	5429	17121	15395782
5	2020-06	5539	5244	16552	14980739
6	2020-07	5851	5533	17512	16003821
7	2020-08	5769	5455	17378	15954931
8	2020-09	5591	5303	16654	14969811
9	2020-10	5778	5441	17228	15799599
10	2020-11	5628	5318	16878	15497095
11	2020-12	5892	5534	17959	16285542
12	2021-01	5689	5371	17229	15522423
13	2021-02	5151	4893	15382	14207789
14	2021-03	5914	5576	17364	16037760

15	2021-04	5516	5254	16498	14955289
16	2021-05	5667	5346	17058	15468549
17	2021-06	5536	5242	16550	15153850
18	2021-07	5612	5279	16711	15399655
19	2021-08	5674	5340	16869	15417951

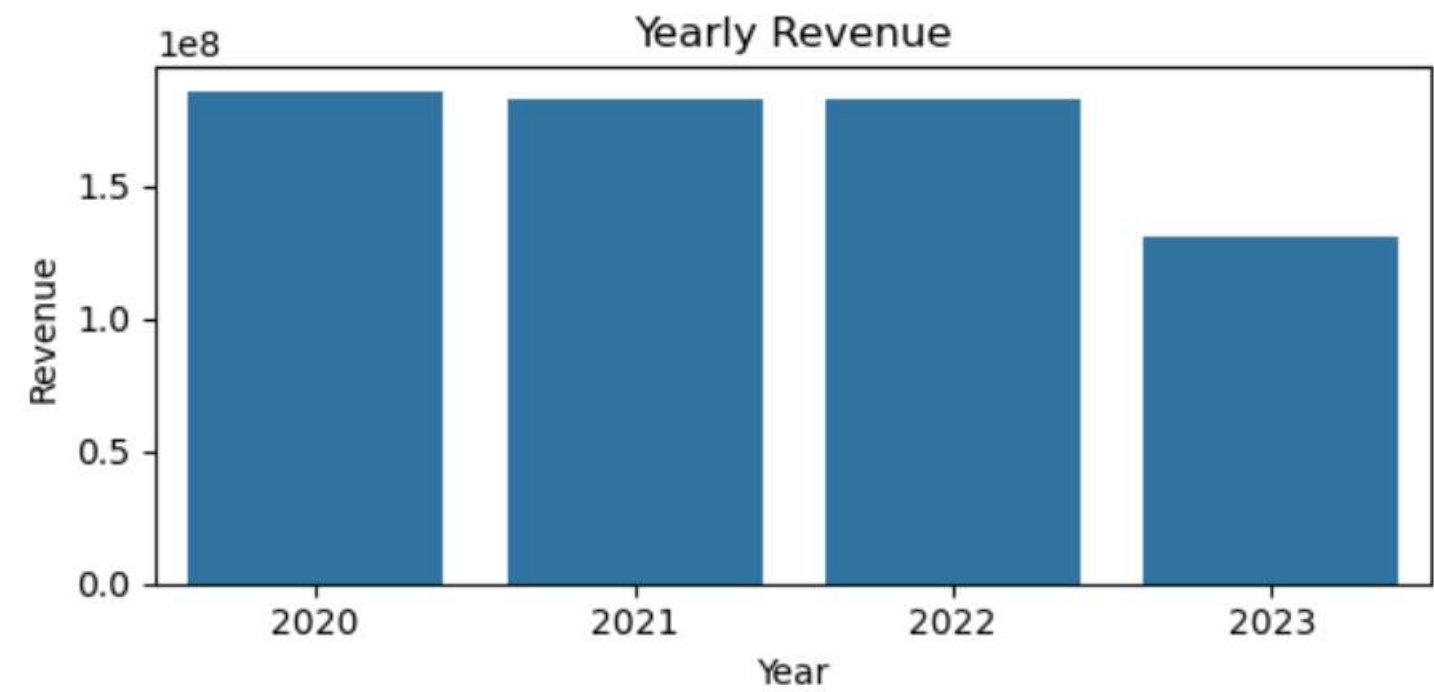


Monthly Transaction Trend



Yearly Purchase Analysis:

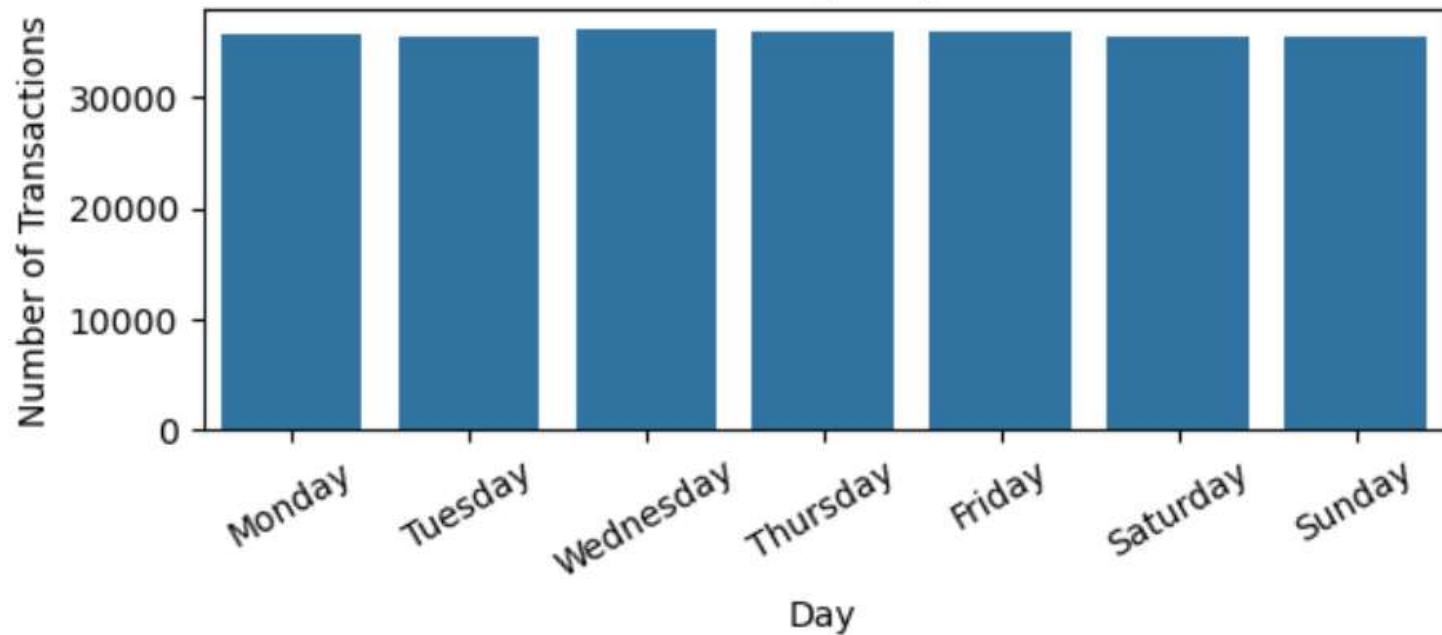
	purchase_year	Transactions	Unique_Customers	Quantity_Sold	Revenue
0	2020	68045	37131	204199	185276174
1	2021	67099	36935	200525	182843403
2	2022	67169	36907	201806	182925502
3	2023	47687	30817	143194	130297604

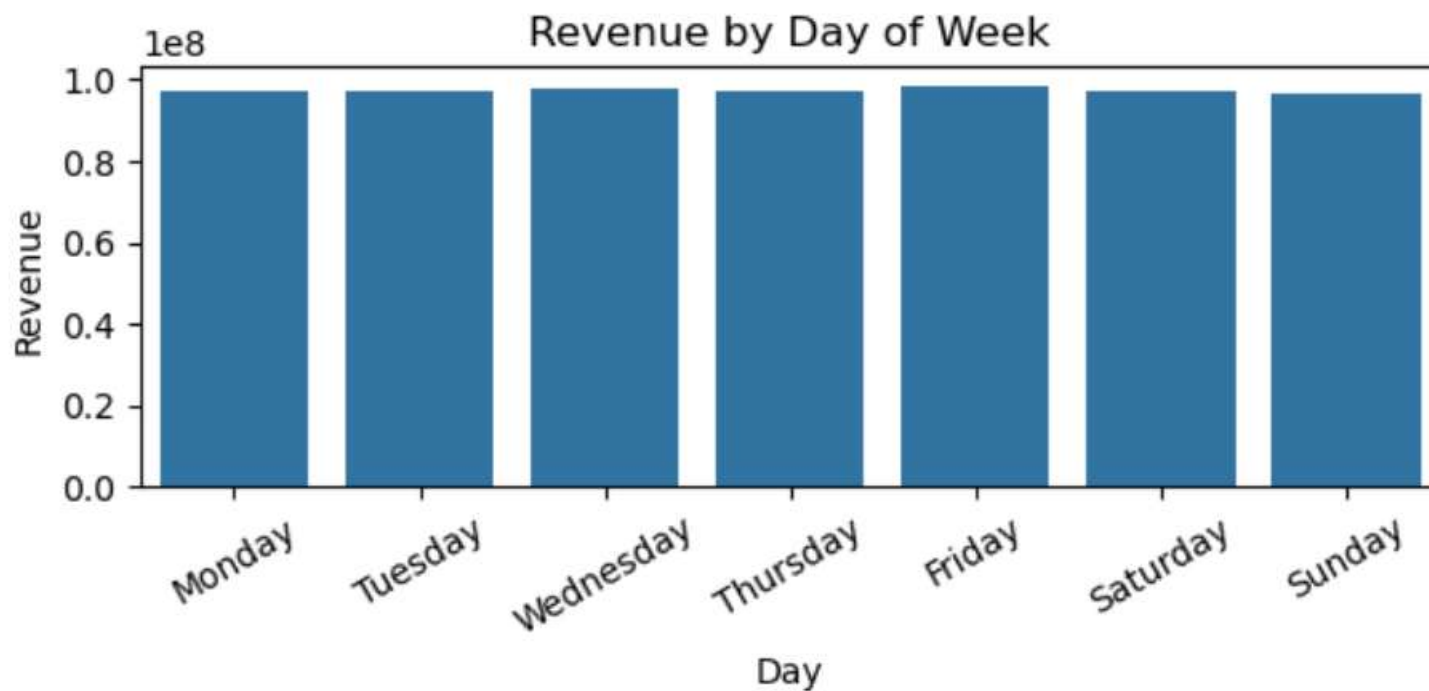


Day-of-Week Purchase Pattern:

	day_of_week	Transactions	Revenue	Quantity_Sold
0	Monday	35710	97322100	107438
1	Tuesday	35553	97273216	106824
2	Wednesday	36048	97916588	108466
3	Thursday	35812	97080211	106918
4	Friday	35890	98208452	107895
5	Saturday	35480	96922086	105937
6	Sunday	35507	96620030	106246

Transactions by Day of Week



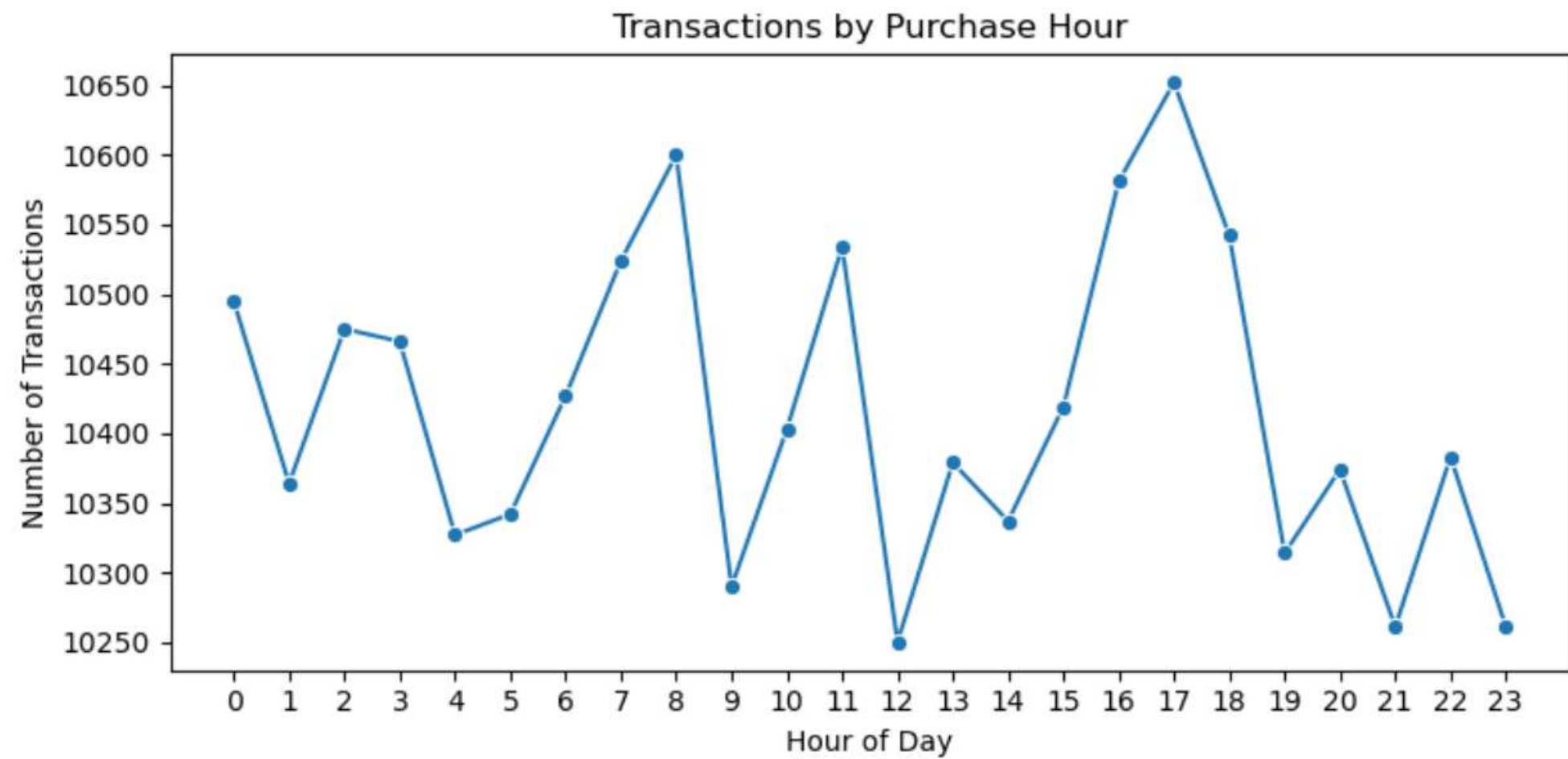


Hourly Purchase Pattern:

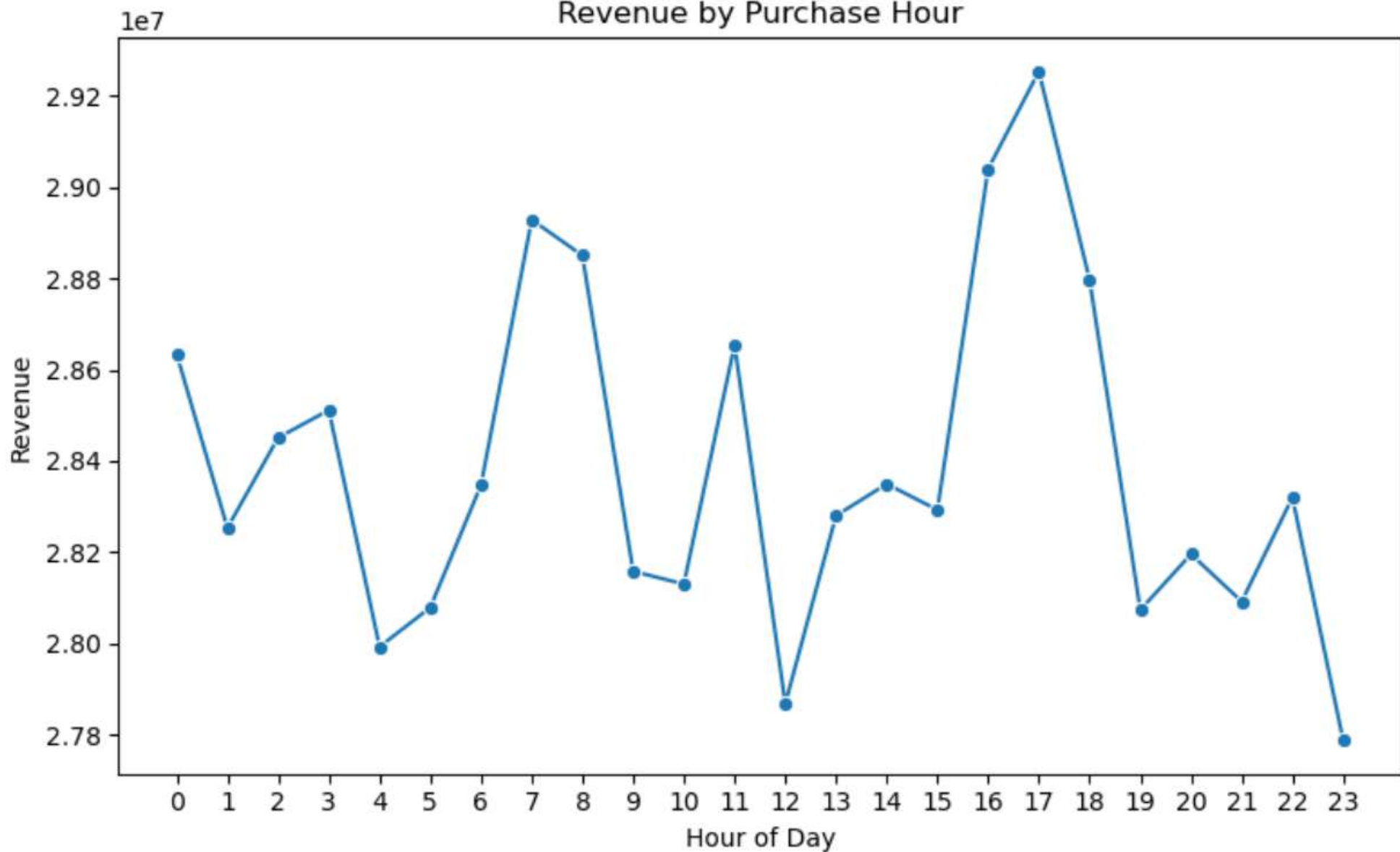
	purchase_hour	Transactions	Revenue	Quantity_Sold
0	0	10495	28633819	31568
1	1	10364	28253618	30988
2	2	10475	28451022	31736
3	3	10466	28511287	31347
4	4	10327	27993258	30997

5	5	10342	28079224	30858
6	6	10427	28347217	31374
7	7	10524	28928170	31597
8	8	10600	28849910	31875
9	9	10290	28158559	30827
10	10	10403	28130549	31167
11	11	10534	28654265	31492
12	12	10250	27868531	30733
13	13	10379	28280321	31387
14	14	10337	28349710	30977
15	15	10419	28293017	31394
16	16	10581	29039442	31705
17	17	10652	29252733	31933
18	18	10542	28796745	31698
19	19	10314	28074080	30710
20	20	10374	28195891	30959
21	21	10261	28092460	30568

22	22	10383	28320286	31223
23	23	10261	27788569	30611

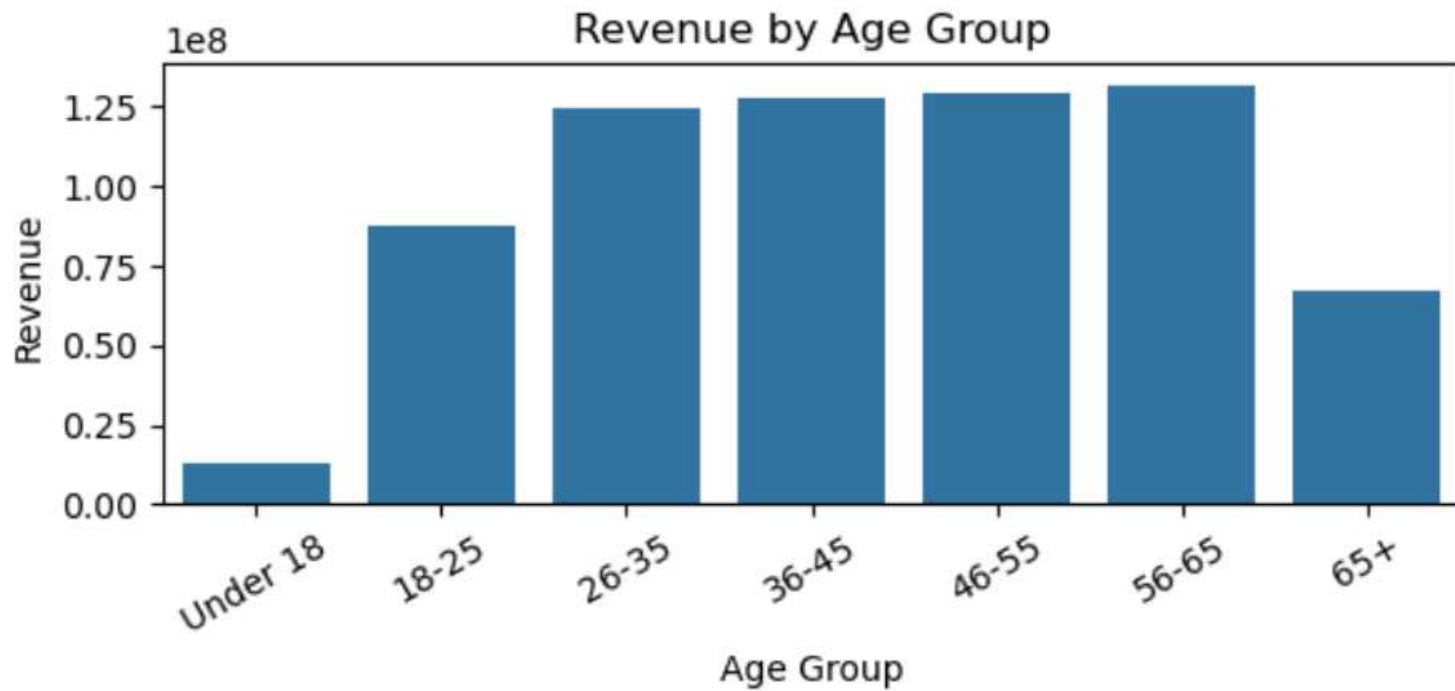


Revenue by Purchase Hour



Age Group Purchase Analysis:

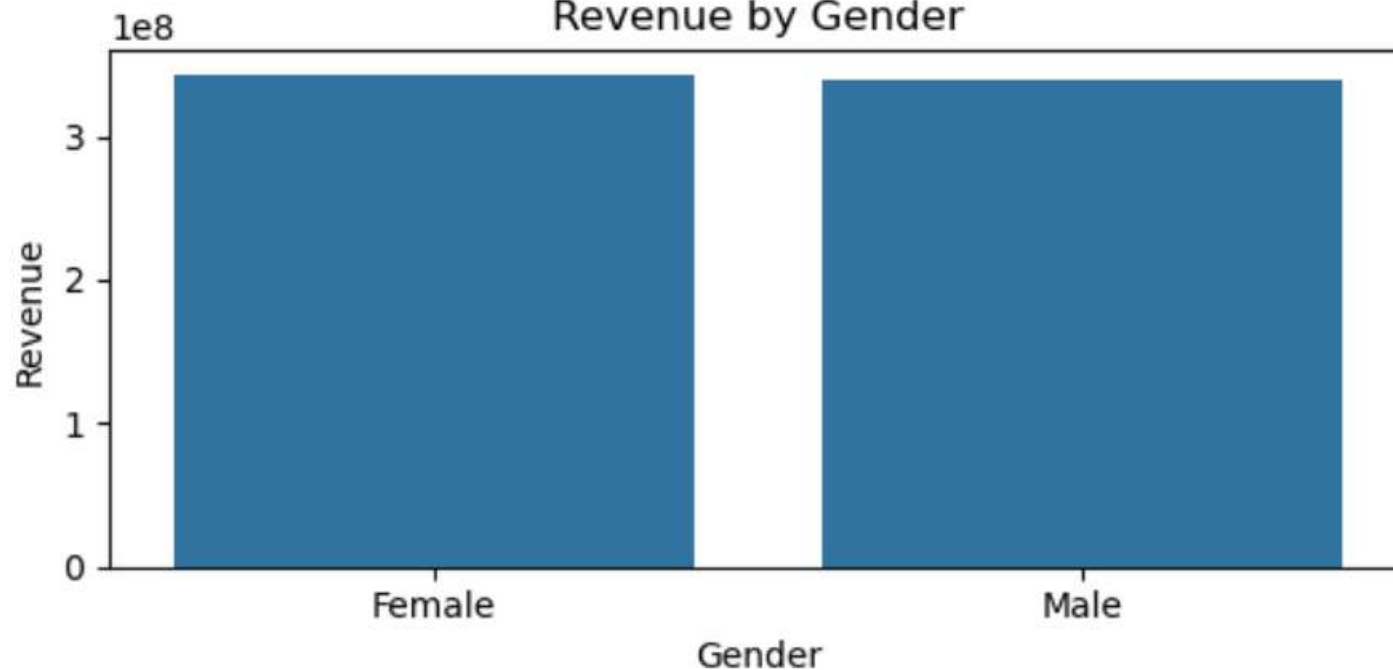
	age_group	Transactions	Unique_Customers	Quantity_Sold	Revenue	Average_Order_Value
0	Under 18	5003	992	15089	13096848	2617.798921
1	18-25	33346	6624	100027	87307757	2618.237780
2	26-35	46906	9334	140249	124568897	2655.713491
3	36-45	47186	9362	141563	127912392	2710.812360
4	46-55	46753	9301	140686	129107480	2761.480119
5	56-65	47120	9338	141405	131874729	2798.699682
6	65+	23686	4722	70705	67474580	2848.711475



Gender-wise Purchase Analysis:

	gender	Transactions	Unique_Customers	Quantity_Sold	Revenue	Average_Order_Value
0	Female	125560	24958	376045	342462421	2727.480256
1	Male	124440	24715	373679	338880262	2723.242221

Revenue by Gender

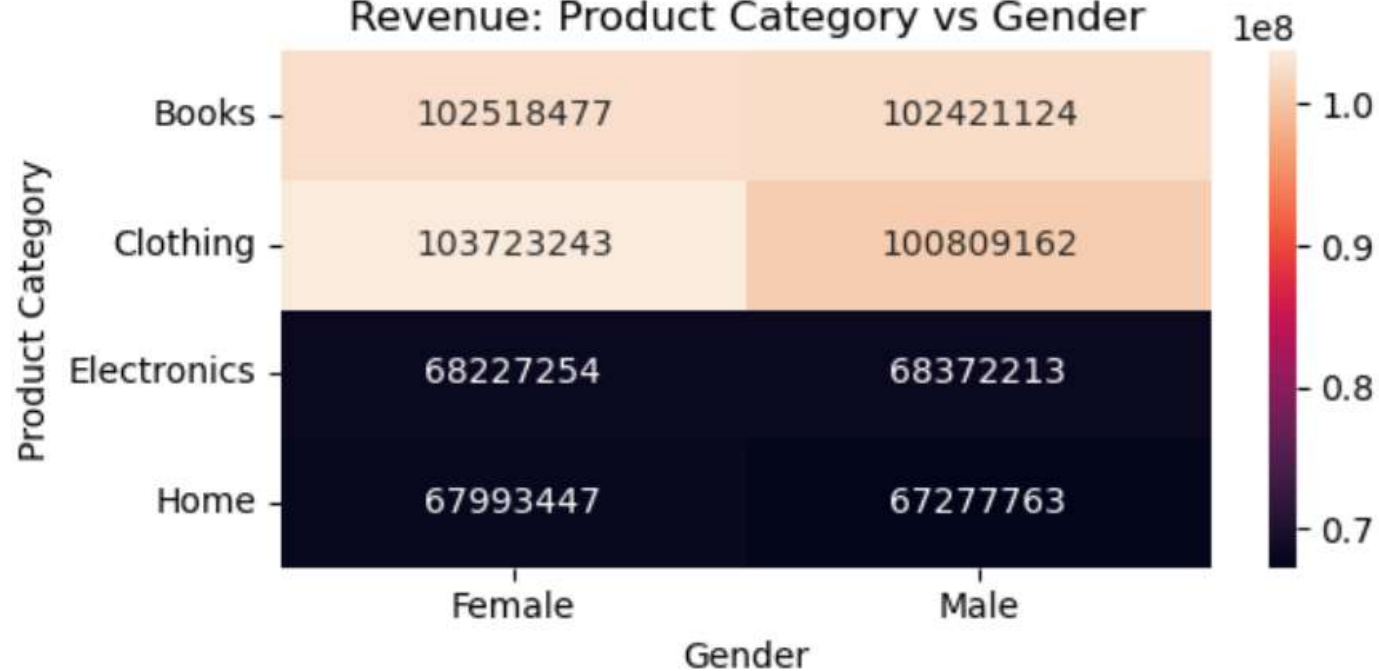


Revenue by Product Category and Gender:

gender	Female	Male
product_category		
Books	102518477	102421124
Clothing	103723243	100809162
Electronics	68227254	68372213

Home 67993447 67277763

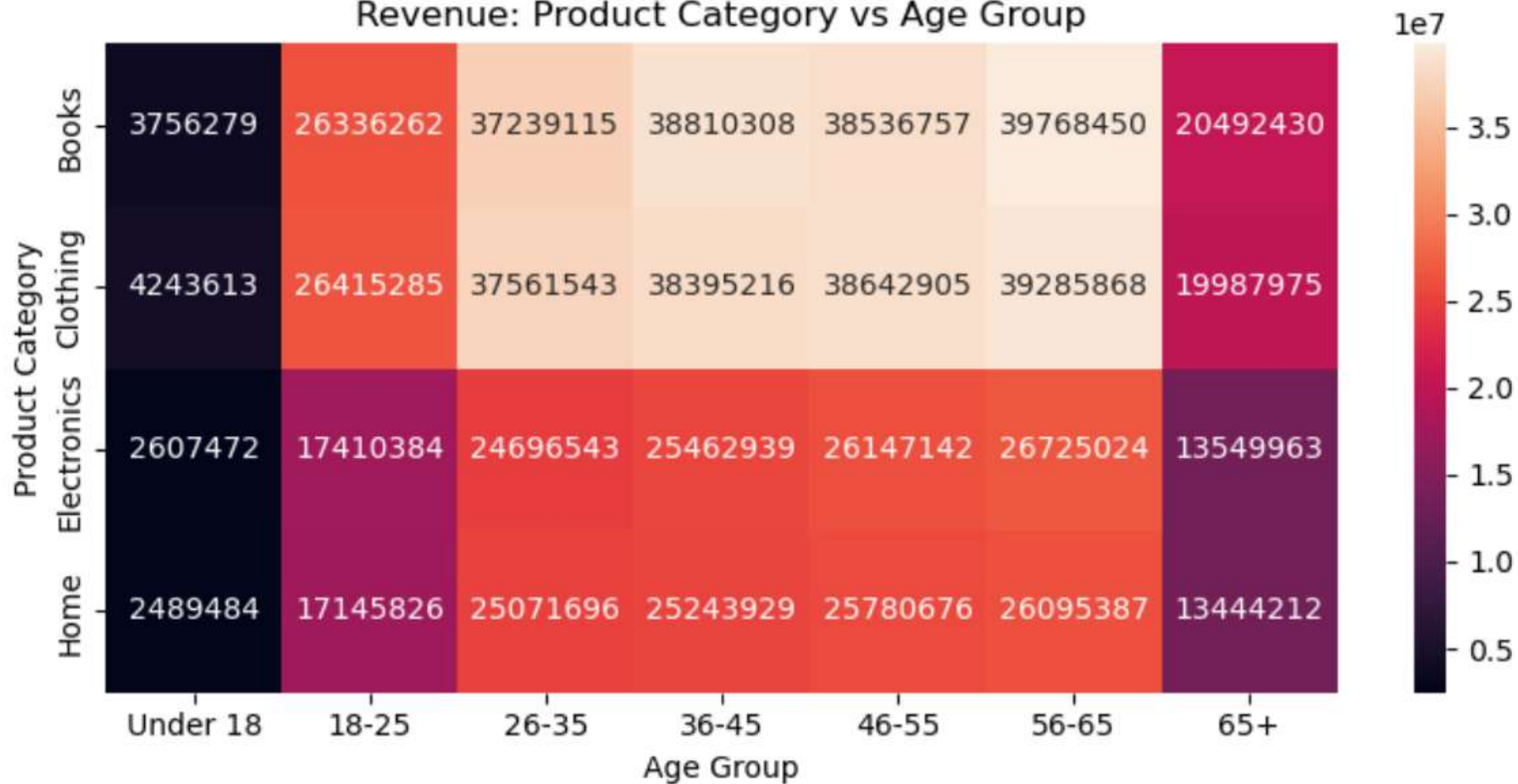
Revenue: Product Category vs Gender



Revenue by Product Category and Age Group:

age_group	Under 18	18-25	26-35	36-45	46-55	56-65	65+
product_category							
Books	3756279	26336262	37239115	38810308	38536757	39768450	20492430
Clothing	4243613	26415285	37561543	38395216	38642905	39285868	19987975
Electronics	2607472	17410384	24696543	25462939	26147142	26725024	13549963
Home	2489484	17145826	25071696	25243929	25780676	26095387	13444212

Revenue: Product Category vs Age Group



Revenue by Product Category and Payment Method:

payment_method Cash Credit Card Crypto PayPal

product_category

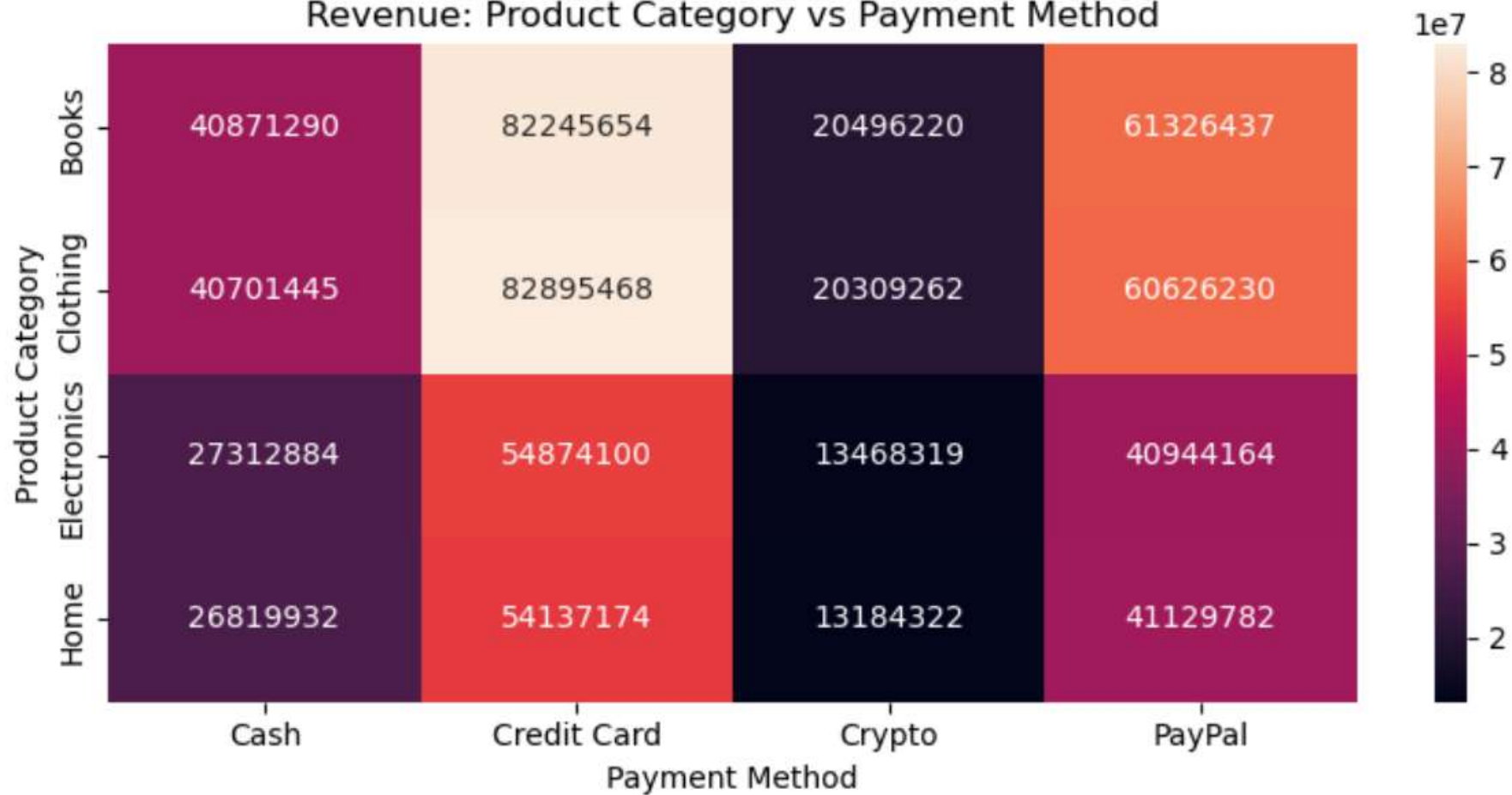
Books 40871290 82245654 20496220 61326437

Clothing 40701445 82895468 20309262 60626230

Electronics 27312884 54874100 13468319 40944164

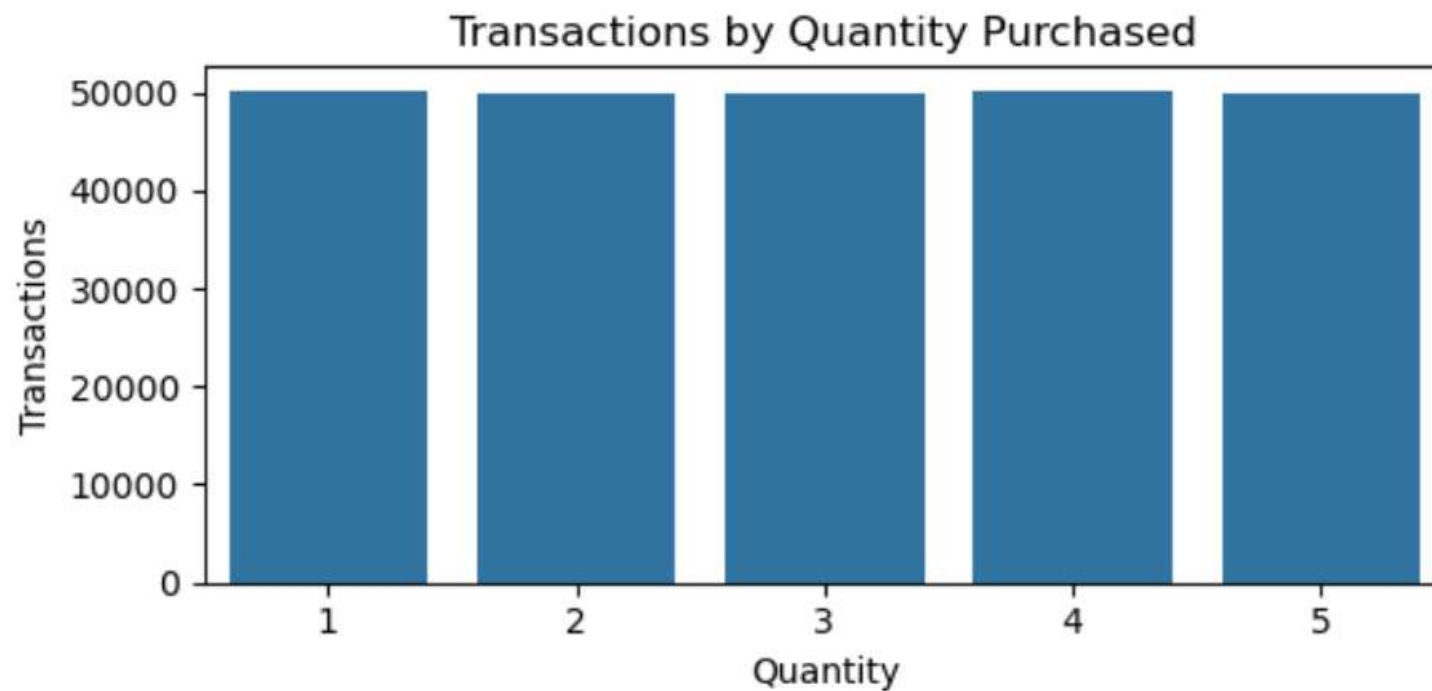
Home 26819932 54137174 13184322 41129782

Revenue: Product Category vs Payment Method



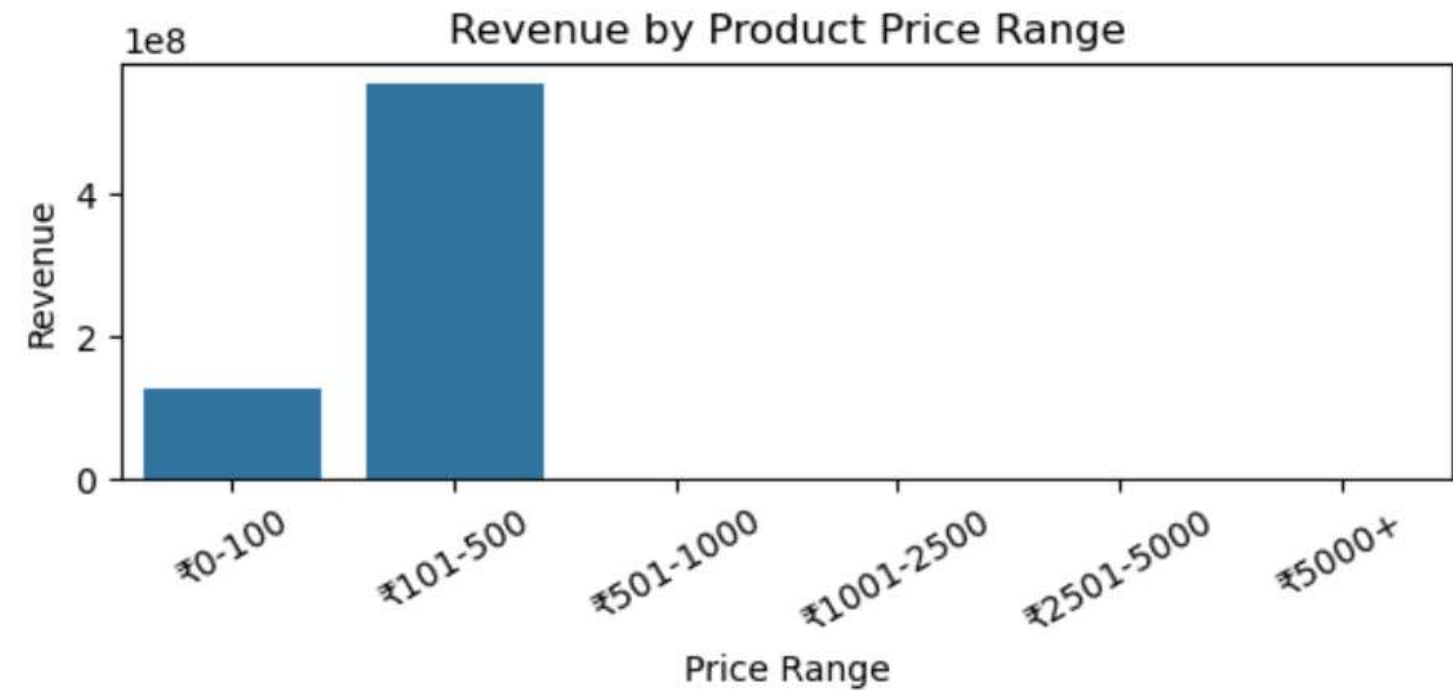
Quantity-wise Purchase Analysis:

	quantity	Transactions	Revenue
0	1	50108	136836518
1	2	50021	135777445
2	3	49872	136029850
3	4	50037	136748269
4	5	49962	135950601



Price Range Analysis:

	price_range	Transactions	Quantity_Sold	Revenue
0	₹0-100	46480	139183	126899832
1	₹101-500	203520	610541	554442851



Return Analysis:

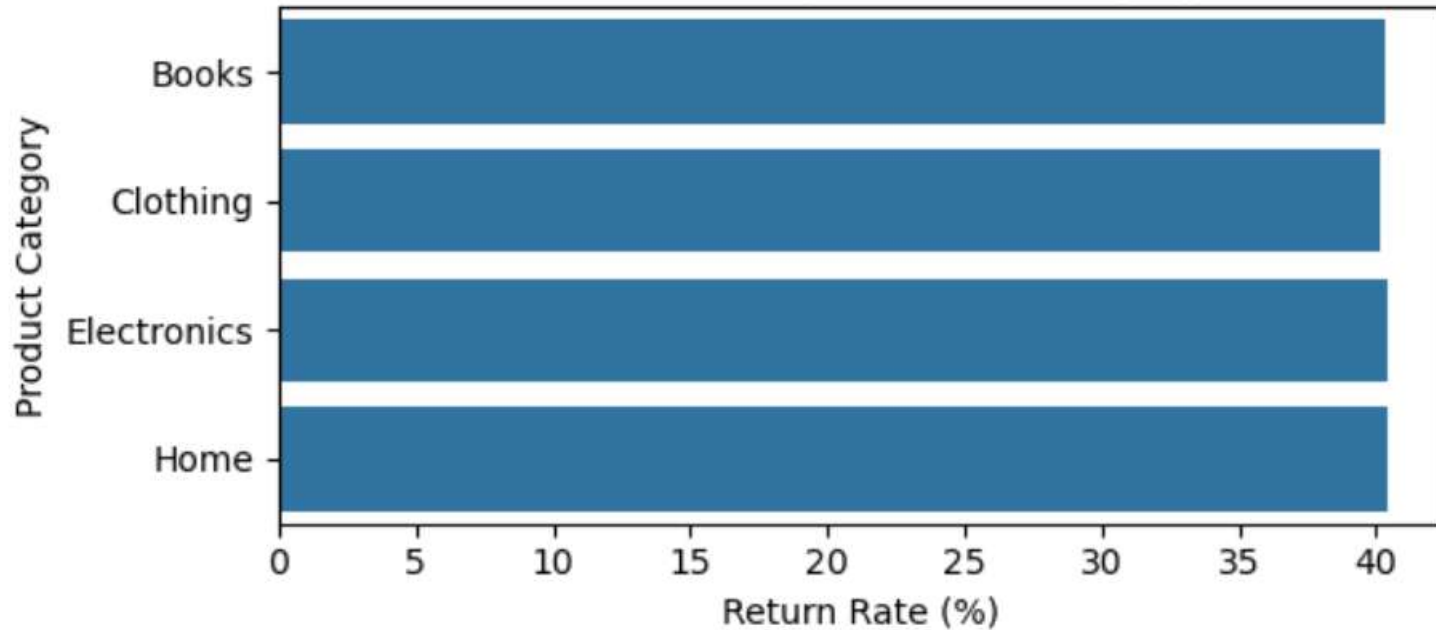
	return_flag	Transactions	Revenue
0	0	149231	406676297
1	1	100769	274666386

Overall Return Rate: 40.31 %

Return Rate by Product Category:

	product_category	Transactions	Returns	Return_Rate
0	Books	74912	30229	40.35
1	Clothing	75052	30122	40.13
2	Electronics	50185	20271	40.39
3	Home	49851	20147	40.41

Return Rate by Product Category



Top 20 Customers by Spending:

	customer_id	customer_name	Total_Spending	Purchase_Frequency	Total_Quantity
36211	36437	Jesse Gray	55339	17	51
39561	39817	Dr. Courtney Foster	51718	17	35
26818	26977	Felicia Escobar	48497	14	26
5809	5844	Maria Owens	46541	12	45
3555	3576	Aaron Harrell	46518	14	38

13718	13802	Jonathan Schroeder	46464	13	41
802	809	Taylor Bishop	45738	15	42
46785	47087	Debbie Lopez	45700	17	51
14313	14400	Patricia Petersen	45334	15	48
20186	20305	Randall Hobbs	45044	14	28
37440	37676	Faith Phillips	44920	13	42
32431	32628	Amanda Thomas	44431	13	41
5379	5409	Douglas Dixon	43287	14	39
4228	4254	John Vazquez	42931	13	31
30654	30838	Cassandra Ingram	42852	14	47
12537	12611	Jennifer Jackson	42804	12	41
15303	15394	Donna Hubbard	42695	12	36
9346	9405	Zachary Gaines	42479	11	34
6739	6786	Peter Williamson	42385	12	37
5222	5252	Adrian Hughes	42137	15	46

Top 20 Customers by Purchase Frequency:

	customer_id	customer_name	Total_Spending	Purchase_Frequency	Total_Quantity
36211	36437	Jesse Gray	55339	17	51
39561	39817	Dr. Courtney Foster	51718	17	35
46785	47087	Debbie Lopez	45700	17	51
14313	14400	Patricia Petersen	45334	15	48
35211	35424	Lindsay Soto	40287	15	51
28675	28852	Chelsea Howell	35178	15	43
12456	12529	Whitney Medina	35671	15	55
5222	5252	Adrian Hughes	42137	15	46
6384	6426	Patricia Elliott	36241	15	38
802	809	Taylor Bishop	45738	15	42
44124	44411	Tammy Hayes	38939	14	37
43764	44049	Jordan Page	39431	14	38
45686	45979	Kelly Smith	37593	14	43
40268	40531	Kristen Leon	37630	14	47
48150	48471	David Duran	40436	14	43

48150	48471	David Duran	40436	14	43
20377	20497	Kim Wood	33934	14	44
19930	20049	Jennifer Green	33583	14	40
27976	28144	Lindsay Morse	41347	14	47
18093	18204	Nicholas Santana	38225	14	39
20186	20305	Randall Hobbs	45044	14	28

Customer × Product Category Matrix:

product_category	Books	Clothing	Electronics	Home
customer_id				
1	3491	0	0	0
2	5169	0	2819	0
3	9150	10570	445	2422
4	0	8715	0	0

5	0	5002	2512	5010
---	---	------	------	------

Most Popular Product Category: Clothing
Highest Revenue Category: Books
Most Used Payment Method: Credit Card
Peak Purchase Day: Wednesday
Peak Purchase Hour: 17
Highest Revenue Month: 2020-12
Revenue: 16285542

=====
STEP 4: PURCHASE PATTERN ANALYSIS COMPLETED
=====

Key Analysis Completed:

- ✓ Overall purchase statistics
- ✓ Product category analysis
- ✓ Payment method analysis
- ✓ Monthly purchase trends
- ✓ Yearly purchase trends
- ✓ Day-of-week purchase patterns
- ✓ Hourly purchase patterns
- ✓ Age group analysis
- ✓ Gender-wise analysis
- ✓ Category × Gender analysis
- ✓ Category × Age analysis
- ✓ Category × Payment analysis
- ✓ Quantity analysis
- ✓ Product price analysis
- ✓ Return analysis
- ✓ Top customer analysis

Important Findings:

Most Popular Category: Clothing

Highest Revenue Category: Books

Most Used Payment Method: Credit Card

Peak Purchase Day: Wednesday

Peak Purchase Hour: 17

Overall Return Rate: 40.31 %

Output files created:

1. category_purchase_analysis.csv
2. payment_method_analysis.csv
3. monthly_purchase_analysis.csv
4. yearly_purchase_analysis.csv
5. day_of_week_analysis.csv
6. hourly_purchase_analysis.csv
7. age_group_purchase_analysis.csv
8. gender_purchase_analysis.csv
9. category_return_analysis.csv
10. top_customers_analysis.csv
11. purchase_pattern_analysis_data.csv